

Krystal Ball — Hotel Bar Inventory Forecasting & Par-Level Recommendation

1. Core Business Problem & Operational Impact

Hotel bars face two competing inventory risks: stockouts of high-demand products and overstocking of slow-moving products. Stockouts can interrupt service and cause potential lost sales, while excessive inventory ties up working capital, consumes storage capacity, and increases exposure to shrinkage, breakage, and spoilage. The project converts transaction-level bottle-balance records into daily demand series for each bar and brand, then uses forecasting and inventory policy calculations to recommend dynamic par levels.

2. Assumptions Made

The baseline assumes a supplier lead time of 2 days, a 95% target service level ($Z = 1.645$), consistent millilitre units, and lost rather than backordered unmet demand. Missing Bar $\times$ Brand $\times$ Date combinations are explicitly represented as zero consumption. Demand variability is estimated from a trailing 28-day window and lead-time standard deviation is approximated as daily standard deviation $\times \sqrt{L}$. The historical evaluation uses a temporal holdout rather than randomized cross-validation.

3. Model Selection & Trade-Offs

The implementation compares a 7-day moving average with a 7-day seasonal-naive forecast. These transparent baselines are appropriate starting points because hotel-bar demand can contain strong weekly patterns and intermittent zero-demand periods. The seasonal-naive method uses the same weekday from the previous week, while the moving average smooths recent demand. More advanced Holt-Winters or tree-based models can be added after establishing these baselines.

4. Performance & Future Improvements

The dataset contains 30,714 transaction records across 5 bars and 10 brands. The inventory conservation check passed for all records. On the final 30-day validation period, the 7-day moving-average baseline produced MAE 1,428.73 ml and WAPE 68.62%; the 7-day seasonal-naive baseline produced MAE 1,662.45 ml and WAPE 79.85%. These are validation-period measurements, not guarantees of future performance. Future improvements include holiday/event calendars, promotions, live POS events, supplier lead-time distributions, minimum-order constraints, and explicit inventory-cost optimization.

5. Real-World Deployment & Production Considerations

A production workflow can run daily: ingest new POS/inventory records, validate data quality, update demand series, generate forecasts, calculate safety stock and par levels, compare current stock with reorder points, and publish recommendations to the bar-manager dashboard. Key failure modes include supplier lead-time variability, holiday and event spikes, promotions, new-product cold starts, missing POS records, and discrepancies caused by waste or breakage. Monitoring should track forecast error, data drift, stockouts, lost volume, and inventory turnover.

Forecast Validation

Model	MAE (ml)	RMSE (ml)	WAPE
7-Day Seasonal Naive	1,662.45	2,989.37	79.85%
7-Day Moving Average	1,428.73	2,226.26	68.62%

Dataset totals: 19,888,597 ml consumed, 19,537,000 ml purchased, and 16,502,416 ml of potential lost demand represented in the synthetic data.